2017 EXAMINATIONS

Lancaster University

Part I (First Year)

SCHOOL OF COMPUTING AND COMMUNICATIONS

SCC.120 Fundamentals of Computer Science (1 hour & 30 Minutes)

- **Answer any THREE out of the four questions.**
- **Use a separate answer book for each question.**

Question 1

1.a

i. Given the sets A={1,2}, and B={3,4}, show the result of A x B.

[2 Marks]

ii. Given the sets A={1,2,3} and B={a,1,c}, show the result of A U B.

[1 Mark]

iii. Given the sets A={1,2,3} and B={a,1,c}, show the result of A ∩ B.

[1 Mark]

iv. The set A={1,2,3} is a subset of B={1,2,3}, but is not a proper subset. Explain why.

[1 Mark]

v. Given the sets A={a,b,c,d} and B={c,d}, use the result of A x B to generate a graph of the results.

[2 Marks]

1.b

i. Explain what we mean by the "domain" and "codomain" of a function.

[2 Marks]

ii. The function f(x)=3x+2x, has the codomain {5,20,40,80}. Derive it's inverse function in terms g(x)=?. Use this to determine the domain. Show any working.

[2 Marks]

iii. Given the domain {1,2,3}, calculate the function f(x)=? that results in the codomain {6,10,14}. Show any working.

[2 Marks]

iv. For the function f(x)=3x-4, derive its recursive form.

[2 Marks]

v. Using the function $f(x)=3x^2$, and the domain {-2,-1,0,1,2}, determine the codomain and show if this function is a bijection. Explain why.

[2 Marks]

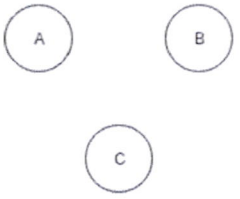
vi. Derive the inverse function for f(x)=2x+5. Show any working.

[2 Marks]

Question 1 continues on next page…

Question 1 continued.

1.c For the questions below, start with the following nodes as the start of your graphs:



i. Draw a graph that demonstrates the transitive property. You do not have to use all the states.

[2 Marks]

ii. Draw a graph that demonstrates the reflexive property. You do not have to use all the states.

[2 Marks]

iii. Draw a graph that demonstrates the symmetric property. You do not have to use all the states.

[2 Marks]

[Total 25 marks]

Question 2

2.a What's the decimal equivalence of 8-bit two's complement binary 11100110?

[2 marks]

2.b Consider the following C code which takes an array of integers, the size of the array, and a target number as arguments.

```
int mystery (int* array, int size, int target) {
        int low = 0;
        int high = size - 1;
        while(low <= high) {
              int mid = (low + high)/2;
              if (array[mid] == target)
                    return mid;
              else if (array[mid] < target)
                    low = mid + 1;
              else high = mid - 1;
        }
        return -1;
}
```

Here we have an array int a[] = {-3, 4, 5, 7, 8, 12, 13, 20};

Give the results of two function calls, (1) mystery (a, 8, 5) and
(2) mystery (a, 5, 12).

[6 marks]

2.c

i. Convert an array with values [22, 12, 2, -10, 30, 22, 35, 6] into a min-heap. Draw the min-heap to show the final result.

[3 marks]

ii. Give the array representation of the min-heap that you have constructed above. Note that the array indices start from 1 in this question.

Index	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Value								

[3 marks]

Question 2 continues on the next page...

Question 2 continued.

2.d We have a structure type called node which is defined as follows:

```
typedef struct _node {
    char token[4];
    struct _node* next;
} node;
```

Consider the following C code:

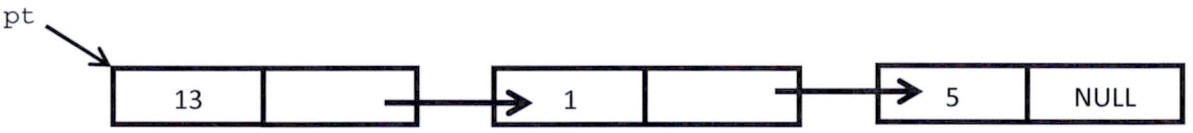
```
1   node* pt = NULL;
2   unsigned int size;
3   node* foo(const char* s) {
4       size = sizeof(node);
5       node *p = malloc(size);
6       strcpy(p->token, s);
7       p->next = pt;
8       pt = p;
9       return p;
10  }
11
12  int main() {
13      foo("SCC");
14      foo("120");
15      foo("Paper");
16      return 0;
17  }
```

i. Assume each character occupies one byte, the word size of the target machine is 4 bytes and the target machine uses 32 bits for memory addressing. What is the value stored in variable `size` after calling function `foo()`? **[2 marks]**

The program above has a bug which may lead to a program crash.

ii. Which line of code in the main() function may crash the program? **[3 marks]**

iii. Assume the bug above has been fixed. Use the "box-and-arrow" notation (an example is shown below) to illustrate what the linked list (that **pt** points to) looks like after executing the main() function.



A general example of a "box-and-arrow" diagram. Your answer would have three boxes with string values.

[6 marks]

[Total 25 Marks]

Question 3

3.a What is the worst case complexity of wellDone? Explain you answer. Its input is an array of integers A.

```
wellDone(int A[]) {
     for (int i=1; i<A.length; i++) {
          for (j=i-1; j>=0 && A[j]> A[i]; j--) {
               A[j+1] = A[j];
          }
          A[j+1] = A[i];
     }
}
```

[4 marks]

3.b What is the worst case complexity class of the following code fragment, where n is the size of the input array? If the input is limited to arrays of size less than 100, what is the complexity then?

```
for (j=10; j<n; j=10*j) {
     print j
}
```

[4 marks]

3.c What is the worst case complexity of the following code fragment? Explain your answer.

```
int a(int n) {
     if(n is 0 or n is 1)
          return n
     else
          return a(n-1) + a(n-2)
}
```

[5 marks]

Question 3 continues on next page…

Question 3 continued.

3.d For each of the following pairs of functions, state which one grows faster asymptotically for input of size n (i.e. is worse in terms of efficiency).

i. $f(n) = 1000n + 53$ and $g(n) = \log_{10}(n)$ **[2 marks]**
ii. $f(n) = n!$ and $g(n) = 2^n + n$ **[2 marks]**
iii. $f(n) = n^3$ and $g(n) = 2^n + n$ **[2 marks]**
iv. $f(n) = \log_2(n)$ and $g(n) =$ square-root(n) **[2 marks]**

3.e Indicate **all** the statements that are true about propositional satisfiability? (Note: More than one statement may be true. No points awarded if you fail to indicate all true statements.)

i. Truth tables are a straightforward way to solve it
ii. It is intractable
iii. It can be reduced to subset-sum in polynomial time
iv. It is well-known that it is not in the complexity class P

[2 marks]

3.f If a problem, say X, is in the complexity class NP, then which of the following statements are true of X?

i. X can be verified in polynomial time
ii. X is a decision problem
iii. If an NP-complete problem can be solved in polynomial time, then X can be solved in polynomial time
iv. X cannot be solved in polynomial time

Indicate **all** true statements (Note: More than one statement may be true. No points awarded if you fail to indicate all true statements.)

[2 marks]

[Total 25 marks]

Question 4

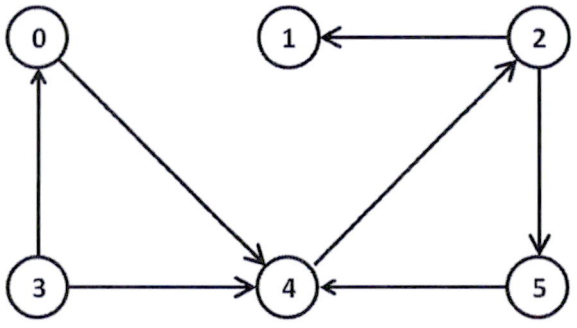
4.a For the "Set" abstract data type, what is the O() complexity to add an element and the O() complexity to remove an element? Explain your answers.

[2 marks]

4.b You are given four numbers (n_1, n_2, n_3, and n_4). They are added in this order into a stack. Then you remove all four of them from the stack. You then take the same four numbers (in the original order) and add them to a queue. Then you remove all four of them from the queue. Give one example of four numbers (i.e. with actual values) where the two ordered sequences (the one removed from the stack and the one removed from the queue) give the same values.

[2 marks]

4.c Consider the directed graph **G** shown below:



i. State whether **G** is non-connected, weakly connected, or strongly connected. Explain your answer.

[1 mark]

ii. How many nodes of **G** have the same in-degree and out-degree? Explain your answer.

[2 marks]

iii. What is the density of **G**? Explain your answer.

[2 marks]

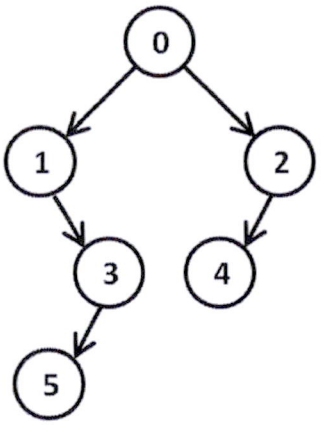
iv. Write down the order in which the nodes are visited during depth-first traversal, starting at node 0. Include the symbol ↑ wherever the traversal is forced to back up to a previous node. You can assume that adjacent nodes are selected in increasing order of their value.

[4 marks]

Question 4 continues on the next page…

Question 4 continued.

4.d This question is about the following tree.



i. What is the sequence of nodes visited by the in-order traversal?

[2 marks]

ii. How many values of "1" are there in the array for the "parent vector" representation of this tree?

[2 marks]

iii. Draw a tree with a post-order traversal sequence of nodes that is the same as the sequence you have from part (i).

[3 marks]

4.e You have been commissioned to re-implement the core infrastructure for the Twitter social network to enable a much more efficient implementation of their API using a more suitable internal design. Out of the five major abstract data types (ADTs) that we have considered (set, stack, queue, graph and tree), which would you use to store key information about Twitter users and "who follows who". Storage for individual messages (tweets) will be kept separately in a different structure so you can ignore that in your design. At a high level, describe your design and why your choice of ADT is best to model users and the one-way 'follow' links between users in Twitter. Also, explain why each of the other ADTs is not appropriate.

[5 marks]

[Total 25 marks]

---End of Paper---